Dallas International School

Test 2

Topics 1.8-2.6-2.7

11th IB - Math AA

For SL and HL Students

EXERCISE 1 For $y = 2x^2 - 8x + 3$:

1. Write it in the form $y = a(x - h)^2 + k$.

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2. State the coordinates of the vertex.

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3. Find the y-intercept.

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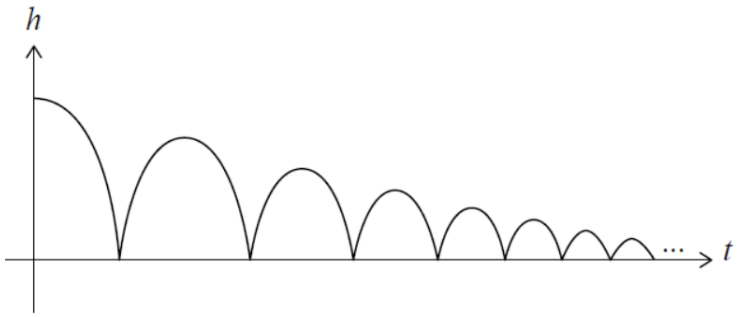
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4. Sketch the graph of the quadratic.

EXERCISE 2 A tennis ball is dropped from a height. After each bounce the maximum height reached by the ball is $\frac{2}{3}$ of its previous maximum height. This can be seen in the diagram below where h, in metres, is the height of a ball after t seconds.



A box contains tennis balls. Each ball satisfies the condition of rebounding to $\frac{2}{3}$ of their previous maximum height. The tennis balls are numbered Ball 1, 2, 3, . . .

Ball 1 is dropped from a height of 10 metres.

1. Find the maximum height of Ball 1 after the 5th bounce.

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2. Find the total distance travelled by Ball 1 immediately before the 5th bounce.

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Let δ be the total distance travelled by any of these balls.

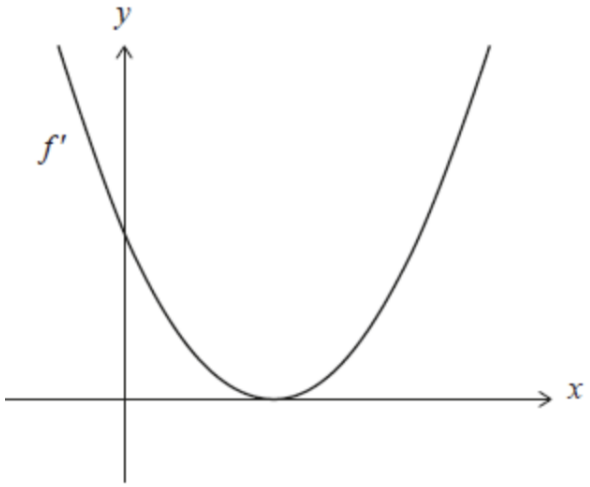
3. A ball is dropped from a height of x metres. Show that $\delta = 5x$ metres.

EXERCISE 3

1. Solve $3m^2 + 5m - 2 = 0$.

2. Hence or otherwise, solve $3 \times 9^x + 5 \times 3^x - 2 = 0$.

EXERCISE 4 A function, f, is given by $f(x) = 3x^2 - 12x + p$, where $p \in \mathbb{R}$. The following diagram shows part of the graph of f.



The graph of f has an axis of symmetry $x = q$.

1. Find the value of q.

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The vertex of the graph of f lies on the x-axis.

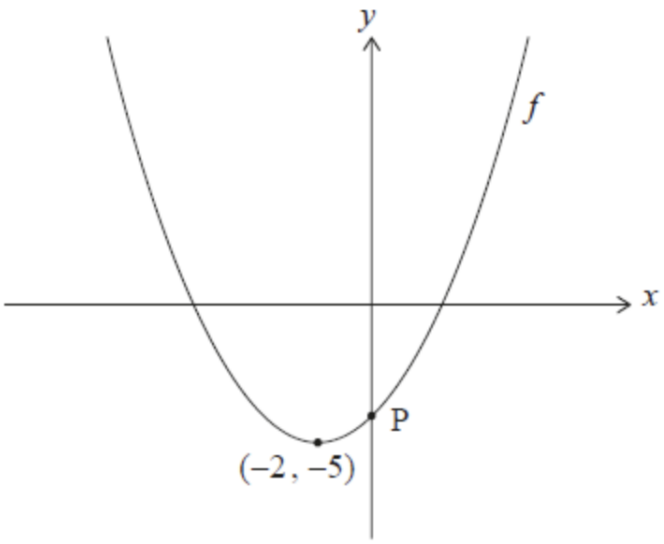
2. a. Write down the value of the discriminant of f.

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b. Hence or otherwise, find the value of p.

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EXERCISE 5 The following diagram shows part of the graph of a quadratic function f.



The vertex of the parabola is $(-2, -5)$ and the y-intercept is at point P.

1. Write down the equation of the axis of symmetry.

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The function can be written in the form $f(x) = \frac{1}{4}(x-h)^2 + k$, where $h, k \in \mathbb{Z}$.

2. a. Write down the values of h and k.

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b. Find the y-coordinate of P.

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